

Supplementary Information

Supplemental Table 1. Means and Standard Deviations of Language Features Analyzed.

Variable	LLM – Means (SD)	High AI Literacy Group – Means (SD)	Low AI Literacy Group – Means (SD)
Personal Pronouns	2.06 (2.04)	2.62 (2.78)	3.01 (2.79)
Impersonal Pronouns	3.40 (1.83)	4.28 (2.36)	4.35 (2.65)
Articles	5.01 (2.59)	7.69 (3.77)	7.17 (3.47)
Conjunctions	7.28 (2.24)	6.11 (2.85)	6.60 (2.83)
Auxiliary Verbs	4.11 (2.25)	6.02 (3.46)	6.28 (3.34)
Prepositions	12.40 (3.01)	14.37 (3.86)	14.59 (4.02)
Adverbs	3.50 (1.96)	4.11 (2.61)	4.46 (2.73)
Negation	0.84 (1.05)	0.77 (1.14)	0.85 (1.11)
Quantity	2.61 (1.88)	3.64 (2.71)	3.69 (3.18)
“I” Words	0.35 (0.60)	0.20 (0.85)	0.31 (1.21)
“We” Words	0.08 (0.35)	0.19 (0.85)	0.29 (0.90)
“You” Words	0.59 (1.30)	0.23 (0.90)	0.21 (0.93)
“They” Words	0.49 (0.75)	0.89 (1.30)	1.07 (1.50)
“She/He” Words	0.44 (1.14)	0.95 (2.26)	0.95 (2.13)

Supplemental Table 2. Linguistic Style Matching between Groups.

Variable	LLM to High AI Literacy Group – Means (SD)	LLM to Low AI Literacy Group – Means (SD)	Low to High AI Literacy Group – Means (SD)
Linguistic Style Matching	0.66 (0.14)	0.67 (0.14)	0.69 (0.13)

Supplemental Table 3. Means and standard errors by group with 95% CI.

Pronoun	Group	Week	M	SE	Lower CI	Upper CI
i	LLM	1	0.314	0.054	0.207	0.42
i	LLM	2	0.287	0.045	0.197	0.376
i	LLM	3	0.352	0.06	0.232	0.471
i	LLM	4	0.344	0.044	0.258	0.43
i	LLM	5	0.341	0.051	0.239	0.442
i	LLM	6	0.474	0.062	0.35	0.597
we	LLM	1	0.102	0.029	0.045	0.159
we	LLM	2	0.056	0.022	0.013	0.099
we	LLM	3	0.069	0.024	0.021	0.117
we	LLM	4	0.092	0.054	-0.014	0.198
we	LLM	5	0.078	0.023	0.032	0.123
we	LLM	6	0.056	0.019	0.018	0.094
you	LLM	1	0.456	0.072	0.313	0.599
you	LLM	2	0.524	0.088	0.35	0.698
you	LLM	3	0.401	0.051	0.301	0.502
you	LLM	4	0.705	0.202	0.306	1.105
you	LLM	5	0.667	0.099	0.471	0.863
you	LLM	6	0.77	0.111	0.55	0.989
they	LLM	1	0.585	0.072	0.443	0.727
they	LLM	2	0.489	0.071	0.349	0.629
they	LLM	3	0.6	0.074	0.454	0.747
they	LLM	4	0.3	0.048	0.205	0.395
they	LLM	5	0.412	0.051	0.31	0.513

they	LLM	6	0.551	0.074	0.404	0.697
shehe	LLM	1	0.897	0.158	0.585	1.209
shehe	LLM	2	0.361	0.097	0.169	0.554
shehe	LLM	3	0.37	0.084	0.204	0.536
shehe	LLM	4	0.427	0.08	0.268	0.585
shehe	LLM	5	0.337	0.062	0.214	0.459
shehe	LLM	6	0.202	0.064	0.075	0.329
ipron	LLM	1	3.921	0.174	3.577	4.265
ipron	LLM	2	3.365	0.165	3.038	3.692
ipron	LLM	3	3.318	0.156	3.01	3.626
ipron	LLM	4	3.02	0.141	2.742	3.298
ipron	LLM	5	3.39	0.14	3.112	3.667
ipron	LLM	6	3.359	0.179	3.003	3.714
i	AI High	1	0.47	0.114	0.246	0.694
i	AI Low	1	0.407	0.118	0.176	0.639
i	AI High	2	0.4	0.084	0.234	0.565
i	AI Low	2	0.359	0.087	0.187	0.53
i	AI High	3	0.33	0.06	0.211	0.448
i	AI Low	3	0.31	0.062	0.188	0.433
i	AI High	4	0.259	0.051	0.159	0.36
i	AI Low	4	0.262	0.051	0.16	0.364
i	AI High	5	0.189	0.063	0.066	0.313
i	AI Low	5	0.213	0.063	0.089	0.337
i	AI High	6	0.119	0.088	-0.054	0.292

i	AI Low	6	0.165	0.089	-0.009	0.339
we	AI High	1	0.15	0.082	-0.011	0.31
we	AI Low	1	0.333	0.085	0.166	0.5
we	AI High	2	0.167	0.06	0.048	0.286
we	AI Low	2	0.292	0.063	0.168	0.415
we	AI High	3	0.184	0.043	0.099	0.269
we	AI Low	3	0.251	0.045	0.163	0.339
we	AI High	4	0.202	0.036	0.13	0.273
we	AI Low	4	0.21	0.037	0.137	0.283
we	AI High	5	0.219	0.045	0.13	0.307
we	AI Low	5	0.169	0.045	0.08	0.258
we	AI High	6	0.236	0.063	0.112	0.36
we	AI Low	6	0.128	0.064	0.003	0.253
you	AI High	1	0.142	0.088	-0.03	0.314
you	AI Low	1	0.042	0.091	-0.136	0.22
you	AI High	2	0.154	0.065	0.027	0.281
you	AI Low	2	0.085	0.067	-0.046	0.217
you	AI High	3	0.165	0.046	0.075	0.256
you	AI Low	3	0.129	0.048	0.035	0.223
you	AI High	4	0.177	0.039	0.1	0.254
you	AI Low	4	0.172	0.039	0.094	0.251
you	AI High	5	0.189	0.048	0.094	0.283
you	AI Low	5	0.216	0.048	0.121	0.311
you	AI High	6	0.2	0.068	0.068	0.333

you	AI Low	6	0.26	0.068	0.126	0.393
they	AI High	1	1.011	0.155	0.706	1.315
they	AI Low	1	1.137	0.161	0.821	1.452
they	AI High	2	1.01	0.114	0.785	1.235
they	AI Low	2	1.082	0.119	0.848	1.315
they	AI High	3	1.009	0.082	0.848	1.17
they	AI Low	3	1.027	0.085	0.86	1.194
they	AI High	4	1.009	0.069	0.873	1.145
they	AI Low	4	0.972	0.07	0.833	1.11
they	AI High	5	1.008	0.085	0.84	1.176
they	AI Low	5	0.917	0.086	0.748	1.085
they	AI High	6	1.007	0.119	0.773	1.242
they	AI Low	6	0.862	0.12	0.625	1.098
shehe	AI High	1	0.259	0.223	-0.18	0.697
shehe	AI Low	1	1.16	0.232	0.705	1.615
shehe	AI High	2	0.496	0.165	0.173	0.82
shehe	AI Low	2	1.046	0.171	0.709	1.382
shehe	AI High	3	0.734	0.117	0.502	0.965
shehe	AI Low	3	0.931	0.122	0.691	1.171
shehe	AI High	4	0.972	0.099	0.776	1.167
shehe	AI Low	4	0.817	0.101	0.618	1.016
shehe	AI High	5	1.209	0.123	0.968	1.451
shehe	AI Low	5	0.703	0.123	0.46	0.945
shehe	AI High	6	1.447	0.172	1.108	1.785

shehe	AI Low	6	0.588	0.174	0.247	0.929
ipron	AI High	1	4.249	0.259	3.739	4.758
ipron	AI Low	1	4.55	0.267	4.025	5.074
ipron	AI High	2	4.219	0.191	3.843	4.594
ipron	AI Low	2	4.469	0.198	4.081	4.857
ipron	AI High	3	4.188	0.136	3.92	4.457
ipron	AI Low	3	4.389	0.141	4.112	4.666
ipron	AI High	4	4.158	0.114	3.932	4.384
ipron	AI Low	4	4.308	0.116	4.078	4.539
ipron	AI High	5	4.128	0.142	3.848	4.408
ipron	AI Low	5	4.228	0.142	3.947	4.508
ipron	AI High	6	4.098	0.2	3.705	4.491
ipron	AI Low	6	4.147	0.2	3.754	4.541

Supplemental Table 4. Main Effect between Groups for Pronouns with students with high AI literacy as reference.

Parameter	β	SE
Self-referential singular pronouns ('i')		
Students with Low AI Literacy	0.12	0.09
LLM	0.15	1.0
Self-referential plural pronouns ('we')		
Students with Low AI Literacy	0.01	0.06
LLM	-0.01	0.02
Second person referential singular pronouns ('you')		
Students with Low AI Literacy	-0.02	0.08
LLM	0.35	0.07

Third person referential plural pronouns ('they')		
Students with Low AI Literacy	0.16	0.12
LLM	-0.40	0.50
Third person referential singular pronouns ('she/he')		
Students with Low AI Literacy	0.08	0.19
LLM	-0.50	0.88
Impersonal Pronouns		
Students with Low AI Literacy	0.14	0.24
LLM	-0.90	1.12

Supplemental Table 5. Main Effect between Groups for Pronouns with Between- and Within-Person Differences. Students with high AI literacy as reference.

Parameter	β	SE
Self-referential singular pronouns ('i')		
Between Person Means	1.10	0.54
Within Person Means	-1.02	0.26
Students with Low AI Literacy	-0.001	0.06
LLM	-0.02	0.05
Self-referential plural pronouns ('we')		
Between Person Means	0.80	0.06
Within Person Means	-0.11	0.03
Students with Low AI Literacy	0.01	0.04
LLM	-0.007	0.03
Second person referential singular pronouns ('you')		
Between Person Means	0.81	0.1

Within Person Means	0.0006	0.03
Students with Low AI Literacy	-0.008	0.08
LLM	0.14	0.08
Third person referential plural pronouns ('they')		
Between Person Means	1.0	0.06
Within Person Means	-0.04	0.03
Students with Low AI Literacy	-0.03	0.08
LLM	-0.04	0.07
Third person referential singular pronouns ('she/he')		
Between Person Means	0.97	0.06
Within Person Means	-0.18	0.03
Students with Low AI Literacy	-0.13	0.12
LLM	-0.05	0.10
Impersonal Pronouns		
Between Person Means	1.03	0.06
Within Person Means	-0.08	0.03
Students with Low AI Literacy	0.15	0.15
LLM	0.14	0.14

Supplemental Table 6. AIC and BIC values for each model: (1) Main effect of groups and time only, (2) Main effect of groups and time with main effect of between and within persons, (3) Main and interaction effects of groups and time with between and within person variables. Model 3 was selected due to lowest AIC and BIC.

Variable	Model	AIC	BIC
i	Group Only Main Effect	3836.49	3863.14
i	Group with Between and Within Persons Main Effect	3591.31	3633.38

i	Group with Between and Within Persons Main and Interaction Effects	3562.07	3614.66
we	Group Only Main Effect	2854.36	2881.1
we	Group with Between and Within Persons Main Effect	2339.58	2381.65
we	Group with Between and Within Persons Main and Interaction Effects	2308.1	2360.69
you	Group Only Main Effect	4792.09	4818.83
you	Group with Between and Within Persons Main Effect	4309.72	4351.8
you	Group with Between and Within Persons Main and Interaction Effects	4282.57	4335.16
they	Group Only Main Effect	4749.19	4775.93
they	Group with Between and Within Persons Main Effect	4264.95	4307.02
they	Group with Between and Within Persons Main and Interaction Effects	4244.55	4297.14
she/he	Group Only Main Effect	6110.14	6136.88
she/he	Group with Between and Within Persons Main Effect	5446.37	5488.44
she/he	Group with Between and Within Persons Main and Interaction Effects	5408.9	5461.49
Impersonal pronouns	Group Only Main Effect	6796.4	6823.14
Impersonal pronouns	Group with Between and Within Persons Main Effect	5920.57	5962.64
Impersonal pronouns	Group with Between and Within Persons Main and Interaction Effects	5900.49	5953.08

Supplemental Table 7. Meta AI Literacy Scale (MAILS) Short-Form (11-point Likert scale)

1. I can use artificial intelligence meaningfully to achieve my everyday goals.
2. I can assess what advantages and disadvantages the use of an artificial intelligence entails.
3. I can tell if I am dealing with an application based on artificial intelligence.
4. I can weigh the consequences of using AI for society.
5. I can design new AI applications.
6. I can program new applications in the field of "artificial intelligence".

7. I can usually solve strenuous and complicated problems related to working with artificial intelligence well.
8. Although there are often new AI applications, I manage to always be "up-to date".
9. I can prevent an AI from influencing me in my everyday decisions.
10. I can handle it when everyday interactions with AI frustrate or frighten me.

Per MAILS scale development, AI literacy was the aggregate score calculated from Items 1 (Detect AI), Item 7 (Apply AI), Item 10 (Understand AI), Item 5 (AI Ethics).